Ethical and Social Issues – Part II

COMP1787 Teaching week 9

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Why learn about Ethics ?

Compliance with the accreditation of our programmes with the British Computer Society

LSEP issues is a ‘current’ topic at the moment (UG final year Project)

It will make you more employable

Why learn about Ethics? Continued..

What of the current trend of using the internet as a way of instigating, managing civil unrest?

In the various demonstrations and riots around the world, the use of IT and the internet have become key tools. Is this good or bad?

There has been a lot of publicity in last few years about lost data (mostly from government offices) – the WikiLeaks case is a good example of this

The social media platforms that are letting fake news items spread. Should they be accountable?

What are the legal, ethical, social, and professional issues that spring from this “problem”?

What are Ethics?

Ethics is the practice of making a principled choice between right and wrong.

Ethical principles are notions of behaviour that are commonly accepted in society (but not always).

Different societies have different notions of what behaviour is acceptable.

When situations involve us at a personal level, we strive to understand the problem, to analyse it and seek a solution - this prevents us from making decisions based on intuition or emotional preference.

An individual's ethical stance may be influenced in many ways.

Ethics standards

Ethical standards are personal or business practices or behaviour which are generally considered acceptable by society.

Acceptable ethics can be described as moral or just and unethical practices immoral or unjust.

Sometimes (often) business practices are within the law but cannot be considered ethical.

Opportunities to be unethical in computing

As IT professionals we have opportunity to abuse power given to us in trust

- Passing sensitive personal information on to a third party

User responsibility

- Lack of awareness or ignoring user needs and build poorly tested systems

Company responsibility

- Abusing the trust given to us by the company

Opportunities to be unethical

- HR – hiring relatives
- Marketing – selling unhealthy products to youth/elderly
- Accounting – misrepresenting truth

Legal Issues vs Ethical Issues

What is unethical is not necessarily illegal

Ethics is based on the recognition of certain human rights.

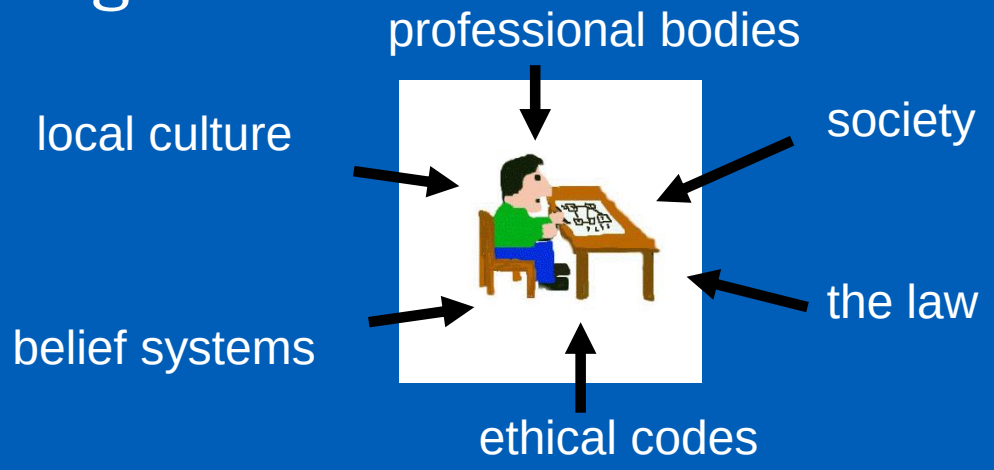
- **Deliberate deception**.
  - Taking credit for someone else's work.
  - Not to be forced to go against their conscience.
  - Sabotaging work of another person
- **Non-work-related use of the Internet**
  - Employees use e-mail and the Web for non-work-related purposes
  - The time employees waste while surfing non-work-related Web sites during working hours is a concern

Laws evolve to mandate ethical behaviour

- Unethical business practices that are now covered by law include:
  - Sending unsolicited e-mail communications
  - Passing sensitive personal information on to a third party
  - Unauthorised accessed to competitor data
  - Copyright infringements

Whose principles?

- When you act ‘ethically’ whose principles are you following?



- Choosing right from wrong
- Choosing right from right

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Ethics in an Information Society

Basic concepts for ethical analysis

Responsibility:

- Accepting the potential costs, duties, and obligations for decisions

Accountability:

- Mechanisms for identifying responsible parties

Liability:

- Permits individuals (and firms) to recover damages done to them

Due process:

- Laws are well known and understood, with an ability to appeal to higher authorities

Privacy

Claim of individuals to be left alone, free from surveillance or interference from other individuals, organizations, or state.

Claim to be able to control information about themselves.

The right of privacy is not absolute. Privacy must be balanced against the needs of society

The public's right to know is superior to the individual's right of privacy

Fair information practices:

- Set of principles governing the collection and use of information
- Basis of most international and local privacy laws
- Based on mutuality of interest between record holder and individual
- Restated and extended by FTC in 1998 to provide guidelines for protecting online privacy
- Used to drive changes in privacy legislation

Fair Information Practices

Principles of Information Systems:

1. Notice/awareness (core principle)
2. Choice/consent (core principle)
3. Access/participation
4. Security
5. Enforcement

Internet Challenges to Privacy:

- Cookies
  - Tiny files downloaded by Web site to visitor's hard drive to help identify visitor's browser and track visits to site
  - Allow Web sites to develop profiles on visitors
- Web beacons/bugs
  - Tiny graphics embedded in e-mail and Web pages to monitor who is reading message
- Spyware
  - Surreptitiously installed on user's computer
  - May transmit user's keystrokes or display unwanted ads
- Google's collection of private data; behavioral targeting

Ethical dilemmas

- Governments will argue that surveillance is a necessary part of protecting a country from its enemies
  - When is ok to undertake electronic surveillance?
  - What controls should be in place?
  - How can you guarantee that surveillance is being undertaken for the right reasons?

Understanding Ethical Challenges and Social Issues (1 of 4)

Key technology trends that raise ethical issues

- Computational Capacity
  - More organizations depend on computer systems for critical operations
- Rapidly declining data storage costs
  - Organizations can easily maintain detailed databases on individuals
- Networking advances and the Internet
  - Copying data from one location to another and accessing personal data from remote locations is much easier

Behavioral Targeting and Your Privacy: You're the Target

Problem: Need to efficiently target online ads

Solutions: Behavioral targeting allows businesses and organizations to more precisely target desired demographics

Google monitors user activity on thousands of sites; businesses monitor own sites to understand customers

Demonstrates technologies role in organizing and distributing information

Illustrates the ethical questions inherent in online information gathering

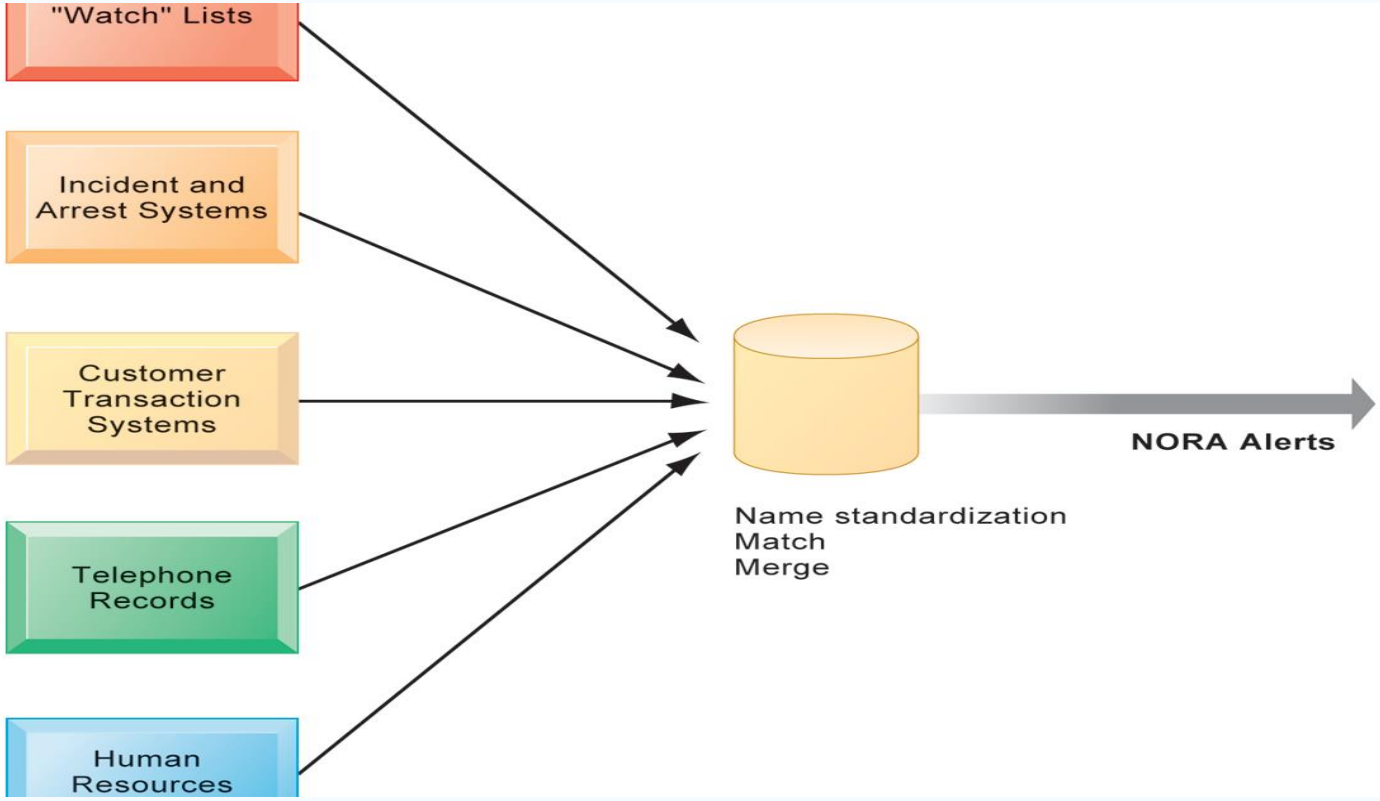
Understanding Ethical Challenges and Social Issues (2 of 4)

- **Biases**
  - Personal biases
  - Algorithm bias
- **Control Problem**
  - Who is making the decision ?
  - Autonomous driven technology
  - High frequency training
- **Privacy**
  - Data how is it trained
  - Toys that have AI – collecting data – how is this being used
  - Face recognition
  - Data aggregator – free apps are collecting data
    - Companies can analyse vast quantities of data gathered on individuals for:
    - Profiling: Combining data from multiple sources to create dossiers of detailed information on individuals
      - Cosmos A
      - Nonobvious relationship awareness (NORA)
    - Combining data from multiple sources to find obscure hidden connections that might help identify criminals or terrorists

Understanding Ethical Challenges and Social Issues (3 of 4)

- NONOBVIOUS RELATIONSHIP AWARENESS (NORA)

- NORA technology can take information about people from disparate sources and find obscure, nonobvious relationships.
- It might discover, for example, that an applicant for a job at a gold store shares a telephone number with a known criminal and issue an alert to the hiring manager.



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Understanding Ethical Challenges and Social Issues (4 of 4)

- Key technology trends that raise ethical issues (*cont.*)
- Power Balance
  - Big monopolies
  - Countries/governments with focused AI strategy
  - Digital divide: The gap between those who have and those who do not have the ability to access electronic technology in general, and the Internet and EC in particular
- Environmental Impact
  - Data centres energy
    - Training in AI 17 x time more carbon emission than an avg American does in a year
  - Is this the right use of this energy
  - Can we use AI to be good for big uses

Computers and Social Interactivity

- Ownership IP
  - Automatic text
  - Deep fake
  - Fake news
  - Art / music – who owns this? But also, who is responsible for this
- Positive aspects of allowing communication across the world
- What are the advantages/disadvantages of an online ‘community’?

Social Issues

- **Education**
  - Virtual universities
  - Home-bound individuals can get degrees
- **Social media**
  - **Chatbots/AI integration**
- **Protecting children**
- **On-line gambling**

- Social implications: TOO MUCH TECHNOLOGY?

What are some of the arguments for and against the use of digital media?

How might the brain be affected by constant digital media usage?

Do you think these arguments outweigh the positives of digital media usage? Why or why not?

What additional concerns are there for children using digital media? Should children under 8 use computers and cell phones? Why or why not?

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A Question

- List all the social, legal and ethical issues that the manager of a sell-side e-commerce web site needs to consider to avoid damaging relationships with users of their site or which may leave the company facing prosecution. You can base your answer on issues which may concern you, your friends or family when you access a web site.

Food for Thought

- **"The Autonomous Car Dilemma"** - A self-driving car is faced with an unavoidable accident that could result in the death of either the car's passenger or a pedestrian.
  - What decision should the car make?
  - Who is responsible for the consequences?

- **"AI Bias in Hiring"** - An employer uses AI to screen job applicants, but the algorithm is biased against certain groups of people.
  - How can the employer ensure fairness and eliminate bias in the hiring process?

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In summary

- Know the difference between Law and Ethics
- The concepts of Ethics has been debated for thousands of years (not new!)
- The evaluation of ethical issues, using frameworks, often results in the making of new laws
- A vast range of issues relate to the technological world and it’s growing!

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Thank you.

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